

MODULAR ANCHORS FOR APÉRY-LIKE COMPANIONS: SECOND SOLUTIONS AS EICHLER INTEGRALS, WITH THE SYMMETRIC-SQUARE THEOREM FOR THE SPORADIC OPERATORS

ABSTRACT. For an Apéry-like sequence $A(n)$ — one of the fifteen sporadic pairs, or a row of the Brown–Zudilin cellular family for $\zeta(5)$ — the *companion problem* asks for a closed formula for the second solution $B(n)$ of its recurrence. We prove an abstract *companion inversion theorem*: if the generating operator is exactly rectified by its nome, $L = C(t) F \theta_q^r F^{-1}$, then the normalized companion is the Eichler-type integral $y_B = F \theta_q^{-r}(t/(CF))$, for all n . We then prove the rectification hypothesis for all fifteen sporadic families: for the nine second-order families it is Frobenius theory; for the six third-order families we prove, by exact operator identities over $\mathbb{Q}(t)$, that each is the symmetric square of an explicit second-order operator $\tilde{D} = \theta^2 - t(2a\theta^2 + a\theta + \frac{b}{2}) + ct^2(\theta + \frac{1}{2})^2$ — a uniform closed form. Consequently the companion formula holds for all n for all fifteen families, including the seven for which we show (computer-assisted, precisely scoped) that no companion exists in the complete tame harmonic ansatz of the programme. Supporting results: the modular identification $F_\zeta = \eta_3^{10}/(\eta_1 \eta_9)^3 = \frac{1}{8}(9E_2(q^9) - E_2(q))$ is proved (Ligozat–Sturm); a first annihilation law for the family ζ ’s summand is proved by a WZ certificate; a new uniform weight-one congruence $A(p) \equiv b_p - b_{\chi_N}(p) \pmod{p}$ survives sixteen held-out primes per family; a curve-blindness theorem for the $\zeta(5)$ top row is stated with a degree-independent reduction (coupling lemma plus a proved Veronese saturation) and finite certified residual computations; and a census of $\sim 5 \times 10^4$ modular pairs finds no sixteenth sporadic pair in its precisely defined search universe. Mirror-type local and arithmetic signatures of the $\zeta(5)$ operator (integral mirror map, integral Yukawa-type datum) and a coaction proposal are collected in a clearly labelled conjectural section. Every statement carries its evidence label; finite computations are never reported as proofs.

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1. INTRODUCTION

1.1. **The companion problem, and what is proved here.** Apéry’s proof of the irrationality of $\zeta(3)$ rests on a pair: the integral solution $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ of his recurrence and the second solution b_n with $b_n/a_n \rightarrow \zeta(3)$. Companion formulas — closed expressions for second solutions

— are the engine of the denominator and congruence arithmetic in the surrounding programme. The classical shape is a binomial sum against a harmonic weight; for seven of the fifteen sporadic Apéry-like families no such formula was known, and the programme’s searches (§5) indicate none exists in the natural finite ansatz.

This paper’s main line is positive and, in its core, proved:

- (1) an abstract *companion inversion theorem* (Theorem 2.4): exact rectification of the operator by its nome implies a closed Eichler-integral formula for the companion, valid for all n ;
- (2) the *rectification theorem for the fifteen* (Theorem 3.2): the nine second-order sporadic operators are rectified by Frobenius theory, and each of the six third-order operators is proved to be the symmetric square of an explicit second-order operator, by exact identities in $\mathbb{Q}(t)$;
- (3) hence the *uniform companion theorem* (Theorem 3.4): for every one of the fifteen sporadic pairs,

$$B(n) = [t^n] F \cdot \theta_q^{-r} \left(\frac{t}{C(t)F} \right) \quad \text{for all } n \geq 0,$$

with r the operator order, q the nome, $F = y_0(t(q))$, and $C = P_r/\sigma^r$ the explicit rectification factor.

Around this proved core we organise, in separate tiers with separate epistemic labels: exact modular results (§4), precisely scoped computer-assisted negatives (§5), arithmetic congruence laws (§6), and a conjectural mirror/coaction programme for the $\zeta(5)$ family (§7).

1.2. Notational dictionary. Throughout, r denotes the *order of the generating differential operator* = the order of the three-term recurrence’s companion matrix +1 in the classical normalisation: $r = 2$ for the nine “weight-2” sporadic rows, $r = 3$ for the six “weight-3” rows. For the families with *identified* parametrizations the modular weight of F is $r - 1$, and the congruence weight w of the programme’s tables equals r ; for the three unidentified Cooper rows ($r = 3$) no modular weight is asserted, and statements grouping them by weight refer to the congruence weight $w = r$ only. We use r exclusively in operator statements; earlier drafts’ floating “ w ” has been retired (referee report, §2). $\theta_x = x d/dx$; t is the recurrence variable, q the nome; $\sigma := \theta_t \log q \in 1 + t\mathbb{Q}[[t]]$.

1.3. Evidence discipline. [THM]/[PROVED]: complete proof (human-readable; computer algebra used only for exact rational identities, which are finite and displayed or scripted). [VERIFIED]: exact rational or modular computation over a stated finite range. [EXCLUDED]: a negative established either by proof or by an exact finite computation over a precisely defined search space, with the space stated. [OPEN]: open. Two standing rules, adopted at the referee’s insistence: finite coefficient agreement is never called an identity, and finite integrality is never called modularity or arithmeticity (§4, §7). A point-by-point response to the referee report, and a log guarding against silent evidential upgrades in future revisions, is maintained at `work/SOL_REVIEW_MODULAR_ANCHORS.md`.

Lemma 1.1 (modular-to-rational transfer). [PROVED] *Let $Mx = v$ be a linear system over $\mathbb{Q}$, p a prime dividing no denominator of the entries of (M, v) , and suppose the reduction $\bar{M}\bar{x} = \bar{v}$ over $\mathbb{F}_p$ is inconsistent. Then $Mx = v$ is inconsistent over $\mathbb{Q}$.*

Proof. A rational solution x has some denominator d ; choosing $p \nmid d$ would reduce it to a solution mod p . In general, if $x \in \mathbb{Q}^m$ solves the system, then for any prime p not dividing the denominators of M, v, x , the reduction solves the reduced system. Inconsistency mod p therefore excludes all rational solutions whose denominators avoid p ; running the computation at two unrelated primes excludes rational solutions except those whose denominators are divisible by both, and the certificate of inconsistency (a left null vector u with $u^T M = 0$, $u^T v \neq 0$ over $\mathbb{F}_p$) lifts: since the entries of M, v are p -integral, any rational solution reduces mod p after clearing its denominator, contradicting $u^T \bar{v} \neq 0$ unless the denominator is divisible by p . As no rational vector has denominator divisible

by every good prime, inconsistency at a single good prime already implies rational inconsistency; we use two primes as an independent guard against implementation error, not as a logical necessity. $\square$

All [EXCLUDED] statements below are of this form: the matrices are exact reductions of rational systems (integer numerators, denominators coprime to the 22-bit primes 4194301, 4194247), so the negatives are theorems over $\mathbb{Q}$ about the stated finite ansatz spaces.

2. THE COMPANION INVERSION THEOREM

Definition 2.1 (nome and rectification). Let L be an order- r operator in θ_t over $\mathbb{Q}(t)$ whose solution space at $t = 0$ has a Frobenius basis $y_0 = 1 + O(t)$, $y_1 = y_0 \log t + g$, $\dots$. The *nome* is $q = t \exp(g/y_0)$, so $\log q = y_1/y_0$; write $t(q)$ for the inverse and $F(q) = y_0(t(q))$. We say L is *exactly rectified* if

$$L = C(t) \cdot F \cdot \theta_q^r \cdot F^{-1} \quad (1)$$

for a rational function C ; equivalently, the kernel of L is $\text{span}\{F, F \log q, \dots, F(\log q)^{r-1}/(r-1)!\}$, i.e. the full Frobenius basis collapses onto powers of the single logarithm $\log q$. For $r \geq 3$ this is strictly stronger than maximal unipotent monodromy (referee report, §1; see Theorem 3.2).

Lemma 2.2 (boundary defect). [PROVED] *Let L come from an R2 or R3 sporadic recurrence and let $y_B = \sum B(n)t^n$ with $B(0) = 0$, $B(1) = 1$ the normalized second solution. Then $L(y_B) = t$ exactly: the coefficients of t^m , $m \neq 1$, vanish because they are instances of the recurrence, and the coefficient of t is $1^r B(1) - (\text{lower terms}) \cdot B(0) = 1$.*

Lemma 2.3 (uniqueness). [PROVED] *A log-free power series y with $y(0) = 0$ and $L(y) = t$ is unique: two such differ by a log-free kernel element vanishing at 0; the log-free part of the kernel is $\mathbb{Q} y_0$ under (1) (all other basis elements carry logarithms), and $y_0(0) = 1 \neq 0$.*

Theorem 2.4 (companion inversion). [THM] *Suppose L is exactly rectified as in (1). Then*

$$y_B = F \cdot \theta_q^{-r} \left(\frac{t}{C(t)F} \right),$$

where θ_q^{-1} is the constant-free antiderivative (well defined since $t/(CF) \in q\mathbb{Q}[[q]]$), and consequently $B(n) = [t^n] F \theta_q^{-r}(t/(CF))$ for all $n \geq 0$.

Proof. The displayed series is log-free, vanishes at $t = 0$, and satisfies $L(\cdot) = C F \theta_q^r (F^{-1} \cdot) = C F \cdot (t/(CF))/1 = t$ by (1). Lemmas 2.2 and 2.3 identify it with y_B . (Intermediate antiderivative constants alter the result by kernel elements with logarithms or by λF ; both are excluded by log-freeness and $y(0) = 0$.) $\square$

3. RECTIFICATION FOR THE FIFTEEN: THE SYMMETRIC-SQUARE THEOREM

3.1. Order two.

Lemma 3.1. [PROVED] *Every order-2 operator with Frobenius basis y_0 , $y_1 = y_0 \log t + g$ is exactly rectified: in the q -coordinate its kernel is $\{F, F \log q\}$ by definition of q , which is the kernel of $F \theta_q^2 F^{-1}$; comparing leading coefficients gives (1) with $C = P_2/\sigma^2$, P_2 the θ_t^2 -coefficient of L . This covers the nine second-order sporadic families (A–F, s_7 , s_{10} , s_{18}), including the $d \neq 0$ Cooper rows.*

3.2. Order three. For an R3 family (a, b, c) (all six have $d = 0$) the generating operator is

$$L = \theta^3 - t(2\theta + 1)(a\theta^2 + a\theta + b) + ct^2(\theta + 1)^3, \quad P_3(t) = 1 - 2at + ct^2.$$

Theorem 3.2 (the six symmetric squares). [PROVED] (*exact identities in $\mathbb{Q}(t)$; work/z5eps/eps57_sym2_all.py*) For each of the six third-order sporadic families, L equals $t^3 P_3(t)$ times the symmetric square of the second-order operator

$$\tilde{D} = \theta^2 - t\left(2a\theta^2 + a\theta + \frac{b}{2}\right) + ct^2\left(\theta + \frac{1}{2}\right)^2,$$

uniformly in the family parameters: writing L in ∂ -form and matching against $\text{Sym}^2(\partial^2 + p\partial + r_*) = \partial^3 + 3p\partial^2 + (2p^2 + p' + 4r_*)\partial + (4pr_* + 2r'_*)$, the forced (p, r_*) agree with $\tilde{D}$'s and the residual $A_0/A_3 - (4pr_* + 2r'_*)$ is identically zero in $\mathbb{Q}(t)$, for $\alpha, \gamma, \delta, \varepsilon, \zeta, \eta$ separately. For example, for γ (Apéry $\zeta(3)$), $p = (2t^2 - 51t + 1)/(t(t^2 - 34t + 1))$, $r_* = (t - 10)/(4t(t^2 - 34t + 1))$.

Corollary 3.3 (rectification for all fifteen). [PROVED] Each of the fifteen sporadic operators is exactly rectified: for $r = 2$ by Lemma 3.1; for $r = 3$ because, with u, v a Frobenius basis of $\tilde{D}$, the kernel of L is $\{u^2, uv, v^2/2\}$ and $v/u = \log q$ (the nome of $\tilde{D}$ equals the nome of L since $y_1/y_0 = uv/u^2$), so the kernel is $\{F, F \log q, \frac{1}{2}F \log^2 q\}$; the leading-coefficient comparison gives $C = P_3/\sigma^3$.

Theorem 3.4 (uniform companion formula, all n). [THM] For every one of the fifteen sporadic pairs, with r the operator order,

$$B(n) = [t^n] F \cdot \theta_q^{-r} \left(\frac{t \sigma^r}{P_r F} \right) \quad \text{for all } n \geq 0.$$

Proof. Theorem 2.4 with Corollary 3.3 and $C = P_r/\sigma^r$. $\square$

Computer-assisted finite verification 3.5. [VERIFIED: exact $\mathbb{Q}$, $n \leq 20$ per family ($n \leq 22$ for ζ , $n \leq 26$ for γ); eps52] The formula of Theorem 3.4 was independently confirmed coefficientwise for all fifteen families; with the theorem now proved, these computations serve as implementation checks.

Remark 3.6. The referee's structural point stands recorded in Definition 2.1: rectification at $r \geq 3$ is strictly stronger than MUM, and a generic MUM operator retains a Yukawa-type coupling in canonical coordinates. What Theorem 3.2 shows is that the sporadic zoo sits entirely in the rectifiable class — order-appropriate in each case (referee report §6) — while the $\zeta(5)$ five-block does not (§7). The half-integral shift $\theta + \frac{1}{2}$ in $\tilde{D}$ is the source of the “half-Apéry” phenomena studied in the companion paper papers_out/half_apery.

4. EXACT MODULAR RESULTS

4.1. The ζ identification is an identity.

Proposition 4.1. [PROVED] (*Ligozat conditions + Sturm bound; eps57, Part 2*) The eta quotient $\eta_3^{10}/(\eta_1\eta_9)^3$ is a holomorphic modular form of weight 2, level 9, trivial character (cusp orders 0, 1, 0 at the cusps 1, $1/3$, ∞ of $\Gamma_0(9)$; $\prod d^{e_d} = 81$ a square), and equals $\frac{1}{8}(9E_2(q^9) - E_2(q))$ in $M_2(\Gamma_0(9))$: the Sturm bound is 2 and agreement holds to q^{40} .

4.2. The first proved annihilation law for ζ 's letters.

Proposition 4.2. [PROVED] (*order-2 Zeilberger certificate with symbolically verified certificate identities, endpoint vanishing, and upward induction; work/z5eps/eps45_zeta_kernel.md, eps45_verify1.py*) For the summand $S_\zeta(n, k, l) = \binom{n}{k}^2 \binom{n}{l} \binom{k}{l} \binom{k+l}{n}$ and all $0 \leq k \leq n$,

$$\sum_l S_\zeta(n, k, l) (H_k^{(1)} - 2H_l^{(1)} + H_{n-l}^{(1)}) = 0.$$

Order 2 is minimal for the certificate (order 1 excluded by residual-checked exhaustion). A second, weight-two annihilation law remains [VERIFIED: exact $\mathbb{Q}$, all cells $n \leq 25$] with a documented

obstruction record (*Gosper and joint order-2–4 solves excluded, also over an expanded alphabet at orders 2–3*) [OPEN]. Under the inner sum the letters $H_l^{(1)}$, $H_{n-l}^{(1)}$ are therefore provably not free — every previous fit for ζ was conducted in a degenerate alphabet.

4.3. Identifications at discovery grade.

Computer-assisted finite verification 4.3 (the dictionary). [VERIFIED: coefficientwise to q^{25} with zero-tail checks; eps51; table in `work/SPORADIC_MODULAR_DICTIONARY.md`] All fifteen nome expansions are integral at scale one to the stated order, and twelve of fifteen (t, F) pairs agree to q^{25} with explicit (generalized-)eta quotients — including δ (level 12: $t = q \eta_1^4 \eta_4^4 \eta_6^{16} / (\eta_2^{16} \eta_3^4 \eta_{12}^4)$, $F = \eta_2^{12} \eta_3 \eta_{12} / (\eta_1^3 \eta_4^3 \eta_6^4)$), apparently the first modular data attached to δ ; η (level 20); **B** (level 36, with the half-integral q -shift anomaly $\sum m e_m = 132 \notin 24\mathbb{Z}$ reflecting the $b(-q)$ twist, and an equivalent untwisted level-9 representative found by the census of §5.3). Cooper’s three are integral but not eta-type in this coordinate [OPEN].

Finite integrality and finite agreement are *discovery* statements: they do not by themselves assert modular parametrisation (referee report §4). The upgrade path is exact operator verification; Proposition 4.1 completes the form-identification step for ζ , and for δ the parallel programme is carried out in `work/DELTA_MODULAR_THEOREM.md`: Ligozat holomorphy of both quotients is unconditional, the eta-side annihilator within the order-3 ansatz is exactly one-dimensional and equals δ ’s operator coefficientwise, and one hypothesis remains (the degree-one property of t_δ on the correct Atkin–Lehner extension) [OPEN].

5. COMPUTER-ASSISTED NEGATIVES, PRECISELY SCOPED

All negatives in this section are of the shape of Lemma 1.1: rational inconsistency of a fully specified finite linear system, certified at two unrelated 22-bit primes with identical ranks.

5.1. The tame harmonic ansatz.

Definition 5.1 (the ansatz). For a family with summand S and congruence weight w , the *complete tame degree- ≤ 3 ansatz* is the finite $\mathbb{Q}$ -vector space of weights $\mathfrak{w} = \sum_j c_j M_j$ where M_j ranges over monomials of total harmonic weight exactly w , of degree ≤ 3 in weight-one letters, in the letters $H_x^{(s)}$ (and, in the twisted variant, $K_\chi^{(s)}(x) = \sum_{j \leq x} \chi(j) j^{-s}$ with χ the conductor-matched character), with arguments x ranging over the family’s complete tame argument set: for ζ : $\{n, k, n-k, l, n-l, k-l, k+l-n\}$; for η : $\{n, k, 2k, 3k, 4k, 5k, n-k, n-2k, n-3k, n-4k, n-5k\}$; for **F**: $\{n, k, n-k, l, k-l, n-l, n-k+l\}$. The companion equation is the linear system $B(n) = \sum_{\text{cells}} S \mathfrak{w}$ over the stated fitting rows.

Theorem 5.2 (exhaustion of the ansatz). [EXCLUDED: two primes, identical ranks; excess 90–173; eps44] *For **F**, ζ , η , the system of Definition 5.1 is inconsistent over $\mathbb{Q}$, pure and twisted (columns/rank: 35/23, 119/75; 140/125, 770/734; 418/418, 2530/2530). Together with the programme’s prior exclusions this exhausts the complete tame degree- ≤ 3 ansatz for all seven conjectural families. The positive control (Franel) re-derives its known companion exactly in the same harness with held-out validation.*

Remark 5.3 (scope). This excludes exactly the stated ansatz. It does not exclude: higher degree in weight-one letters, nested harmonic sums, rational-function coefficients $c_j(n, k)$, other hypergeometric representations of the same $A(n)$, or telescoping/indefinite-sum extensions (referee report §3). The claim “no harmonic companion exists” does not appear in this paper; Theorem 3.4 makes the point that matters — the companion *does* exist, in modular letters, for all fifteen — and Proposition 4.2 explains part of the ansatz failure (degenerate alphabets).

5.2. Curve-blindness for the $\zeta(5)$ top row.

Definition 5.4 (curve atoms, pinning, boxes). Write the Brown–Zudilin cell $T = \prod_L (x_L!)^{p_L}$ over the nine letters $x_L \in \{n, k, l, n \pm k, n \pm l, k + l, n + k + l\}$, $p_L = (1, -3, -3, 1, 1, -2, -2, -1, 1)$. A *curve atom of degree D* is $u = (u_1, \dots, u_D)$, $u_i \in \mathbb{Z}^3$, denoting the Γ -continued cell $T(n + \alpha(\varepsilon), k + \beta(\varepsilon), l + \gamma(\varepsilon))$, $(\alpha, \beta, \gamma) = \sum_i u_i \varepsilon^i$; per letter the shift is $s_L = \sum_i d_{i,L} \varepsilon^i$ with $d_{i,L}$ the letter’s linear form on u_i . *Pinning* of a $\mathbb{Q}$ -combination means: the ε^1 and ε^2 rows vanish and the ε^3 row lies in $\mathbb{Q}\mathbb{Q} \oplus \mathbb{Q}\hat{P}$, imposed separately in every ζ -graded component (variant A adds the analogous ε^4 condition). The *boxes* are $u_1 \in \{-2..2\}^3$, $u_2 \in \{-1, 0, 1\}^3$, $u_3 \in \{0, \pm e_i\}$.

Lemma 5.5 (coupling; degree-independent). [PROVED] *For any polynomial curve atom of any degree, the top-graded (harmonic weight = ε -order r) content of $[\varepsilon^r]$ depends only on the first-order direction u_1 , through the pure powers $d_1^{\otimes r}$; all higher curve coefficients feed only lower weights. Hence the top-grade question for all polynomial curves reduces to the span of the degree- r Veronese images of first-order directions.*

Lemma 5.6 (Veronese saturation). [PROVED] (exact rank computation; `eps57_heldout.py`, Part B) *The directions of the box $\{-2..2\}^3$ have degree- d Veronese images of full rank $\dim \text{Sym}^d(\mathbb{Q}^3)$ for $d = 2, 3, 4, 5$ (6, 10, 15, 21). Hence the finite computations below certify the top-graded component for every first-order direction in $\mathbb{Z}^3$, and so, by Lemma 5.5, for every polynomial curve of every degree.*

Theorem 5.7 (curve-blindness). *The top-graded ε^5 content of any pinned $\mathbb{Q}$ -combination of polynomial-curve atoms (any degree, any directions) has vanishing P_n -component; and on the boxes of Definition 5.4 the full statement holds: the entire ε^5 row of a pinned combination is confined to $\mathbb{Q}\mathbb{Q} \oplus \mathbb{Q}\hat{P}$, in the rational part and every ζ -graded component [EXCLUDED: two primes, identical ranks 53/36 (linear), 88/59 (quadratic), 80/54 (cubic); `eps41`, `eps42`]. The ε^3 coefficient meanwhile realises $\hat{P}$ freely. The top-grade clause is unconditional by Lemmas 5.5–5.6; the sub-top clauses are certified on the stated boxes, where the pinning constraints (which involve u_2, u_3) live.*

5.3. The census.

Definition 5.8 (search universe). The census banks are: *hauptmodul* candidates $t = q \prod_m (1 - q^{mk})^{e_m}$ with $\sum e_m = 0$, $\sum m e_m = 24$; form candidates $F = \prod_m (1 - q^{mk})^{e_m}$ with $\sum e_m = 2(r - 1)$, $\sum m e_m = 0$, $r - 1 \in \{1, 2\}$; Eisenstein members $(d_2 E_2(q^{d_2}) - d_1 E_2(q^{d_1})) / (d_2 - d_1)$ and odd-character weight-one series $E_1(\chi_{-3}), E_1(\chi_{-4})$ at rescalings; levels $\{2, \dots, 10, 12, 13, 16, 18, 20, 25\}$; exponent caps $\Sigma|e| \leq 26$ (pass 1) and ≤ 60 (pass 2) with the stated candidate-count caps; and both $(-q)$ -twists of every pair. Two hits are *equivalent* if they agree under $t \mapsto \lambda t$ and $t \mapsto -t$, detected by the invariants a_k/a_1^k ; hit classes are these equivalence classes.

Computer-assisted census 5.9. [VERIFIED: $\sim 5 \times 10^4$ pairs; all 91 hit classes re-derived exactly at series order 46; `eps55`] Within the universe of Definition 5.8: the reverse pipeline reproduces the exact recurrences of all ten identified families from their (t, F) data alone (10/10); and every 3-term hit with integral solution is either one of the fifteen or a classical degenerate (hypergeometric $c = 0$, interleaved/2-step $a = b = 0$, or the double-root case $a^2 - 4c = 0$). *No sixteenth sporadic pair exists in this universe.* The four uncovered directions (generalized-eta/ Γ_1 banks, Fricke *hauptmoduls*, order-4 territory, levels > 25) are stated in `work/MODULAR_FACTORY.md`.

6. CONGRUENCE LAWS

Computer-assisted finite verification 6.1 (weight dichotomy). [VERIFIED: $p \leq 23$ per family; `eps51`] For all nine $r = 3$ -adjacent (modular weight 2) families, $A(p) \equiv b_p \pmod{p}$ with $b_p = [q^p]F$, at every tested prime — including the unidentified Cooper three and the χ -twisted ζ, η .

Proposition 6.2 (nebentypus compatibility; a uniform weight-one law). [VERIFIED: fitted on $p \leq 23$; passes all sixteen held-out primes $29 \leq p \leq 97$ for each of the six families below — the five displayed plus **D** with its separate rule; `eps57_heldout.py`] For the weight-one families,

$$A(p) \equiv b_p - b \cdot \chi_N(p) \pmod{p} \quad (\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{E}, \mathbf{F}),$$

with b the recurrence's constant parameter (2, 3, 3, 4, 6) and χ_N the quadratic character of the level (χ_{-3} for levels 6, 12, 36 — including Franel, whose family character is trivial; χ_{-4} for level 8); and for **D**, $A(p) - b_p \equiv 2(p \bmod 5) - 5 \pmod{p}$. For ζ , the programme's twisted Lucas law holds with twist exponent $e = 1$ equal to the nebentypus value $\chi_{-3}(p)$ at the discriminating primes.

Conjecture 6.3 (ASD mechanism). The laws above are the mod- p shadows of Atkin–Swinnerton-Dyer congruence structure for the identified forms (unit-root/formal-group statements); in particular the weight-one law should follow from the Eisenstein structure of the identified weight-one forms. Establishing this for ζ — where Proposition 4.1 supplies the form as a theorem — is the recommended first target.

7. CONJECTURAL TIER: MIRROR-TYPE SIGNATURES OF THE $\zeta(5)$ OPERATOR, AND A COACTION PROPOSAL

Everything in this section is either an exact local computation about the Brown–Zudilin operator or an explicitly labelled conjecture; no geometric origin, Hodge-theoretic, or motivic claim is made.

7.1. Local signatures.

Proposition 7.1. [PROVED] (exact indicial computation; `eps50`) The order-9 generating operator of the L_{BZ} recurrence has $t = 0$ exponents $0^{(5)}$, $\frac{1}{2}$, and the three roots of an irrational cubic: a five-dimensional unipotent local block.

Computer-assisted finite verification 7.2. [VERIFIED: q^{26} (mirror map), q^{32} (Yukawa datum), scale one; `eps50`, `eps53`] The block's mirror map and structure series are integral: $t(q) = q - 94q^2 - 591q^3 - \dots \in \mathbb{Z}[[q]]$ to the stated order; $F(q) = 1 + 21q + 1015q^2 + \dots \in \mathbb{Z}[[q]]$; and, in the exact q -normal form of the block (defined below), the Yukawa-type datum $K(q) = \theta_q^2(\hat{g}_2/F) = 87q + 33895q^2 + \dots \in \mathbb{Z}[[q]]$, with $K_1 = 87 = 4P_1$.

Withdrawn as evidence: earlier drafts listed the root-integrality observations $Q(4t)^{1/2}, Q(8t)^{1/4} \in \mathbb{Z}[[t]]$ here. These are *universal* for integer series ($\sqrt{f(4t)} \in \mathbb{Z}[[t]]$ for any $f \in 1 + t\mathbb{Z}[[t]]$; see `papers_out/half_apery`, Lemma) and carry no arithmetic information about this family; the correction is logged in `work/Z5_MODULARITY_PROBE.md`.

Definition 7.3 (q -normal form and the datum K). In the nome coordinate of the block, conjugation by F brings the five-dimensional local system to the form

$$\theta_q^5 + K_3(q) \theta_q^3 + K_2(q) \theta_q^2 + K_1(q) \theta_q + K_0(q),$$

with $\hat{g}_1 \equiv 0$ (the mirror-map normalisation, verified exactly) and $\hat{g}_j$ the successive Frobenius tails; $K := \theta_q^2(\hat{g}_2/F)$ is the first nontrivial structure series. $K \neq 0$ is precisely the obstruction to the bare inversion of Theorem 2.4 at $r = 5$: the variation-of-parameters inverse then involves K explicitly. The full tower $\hat{g}_3, \hat{g}_4$ and the invariance of K under the remaining gauge freedom are [OPEN] (referee report §10); until they are supplied, K 's status is: an exactly computed series in a fixed, scripted normalisation (`eps53`).

Computer-assisted finite verification 7.4 (sharp negatives). [EXCLUDED: as stated; `eps53`] (i) The naive Dwork congruence $F(q) \equiv F_{<p}(q)F(q^p) \pmod{p}$ fails at $p = 5, 7$ (three Frobenius slopes; the slope-refined form is the open target). (ii) The naive elliptic-style row avatars $\theta_q^3(\hat{P}/Q), \theta_q^5(P/Q)$ have unboundedly growing denominators — calibrated against the same pipeline recovering Beukers'

weight-4 Eisenstein combination $6E^{(1)} - 168E^{(2)} + 378E^{(3)} - 216E^{(6)}$ exactly for Apéry (q^{32}). (iii) No order-5 right factor of t -degree ≤ 20 splits the block off globally (one 22-bit prime, series order 64).

Conjecture 7.5 (mirror form of the bridge). *In the block's q -normal-form coordinates the companion rows $\hat{P}_n, P_n$ are coefficient sequences of Eichler-type objects built from F, K , and the higher structure series, and the programme's open compact-formula identity for P_n is the t -shadow of a q -side identity in those data.*

7.2. A coaction proposal, in one display.

Conjecture 7.6. *For the motivic period I^m of a cellular family with coaction $\Delta I^m = \sum_r I_r^{\text{dR}} \otimes I_r^m$, the r -th ε -jet of the family computes I_r^{dR} ; in particular companions are first coaction components, and the twist behaviour of Proposition 6.2 is the Galois equivariance of Δ .*

No comparison morphism is constructed here; the conjecture is recorded because it correctly anticipated the one-step (unipotent) shape of the first-variation anchor equation of the companion paper, and it makes the falsifiable prediction that second variations anchor in $\text{span}\{Q, \dot{Q}\}$ only.

8. OPEN PROBLEMS

- P1.** Complete δ 's parametrisation proof (the remaining Atkin–Lehner degree-one hypothesis), and prove the remaining eleven identifications by the Ligozat–Sturm route of Proposition 4.1.
- P2.** Prove the weight-one law of Proposition 6.2 from the Eisenstein structure of the identified forms; then the ASD mechanism (Conjecture 6.3) for ζ .
- P3.** Certify the second ζ annihilation law (Proposition 4.2); mine the quotient alphabet.
- P4.** Supply the $\hat{g}_3, \hat{g}_4$ tower and the gauge-invariance of K (Definition 7.3); run the slope-refined Dwork test; derive the K -corrected inversion at $r = 5$ and test Conjecture 7.5 on the *proved* middle row first.
- P5.** Extend the census (Definition 5.8) in its four uncovered directions; any of them could still contain a sixteenth pair.
- P6.** Identify Cooper's three parametrizations (Atkin–Lehner hauptmoduls).

9. PROVENANCE AND DATA

Scripts in `work/z5eps/` with logs: `eps41-eps44` (negatives), `eps45` (the proved annihilation law), `eps48-eps53` (nome instrument, identifications, companions, mirror probes), `eps54` (the discriminating jet experiments), `eps55` (census), `eps56` (half-Apéry), `eps57_sym2_all.py` and `eps57_heldout.py` (the symmetric-square identities, the ζ Sturm proof, held-out primes, Veronese saturation). Memos as in the header. All exact arithmetic is `Fraction`-based; every modular negative is certified at two unrelated primes (Lemma 1.1); every positive control is stated alongside the negatives it guards. The referee report and the point-by-point response — including a standing log against silent evidential upgrades — are at `work/SOL_REVIEW_MODULAR_ANCHORS.md`.

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